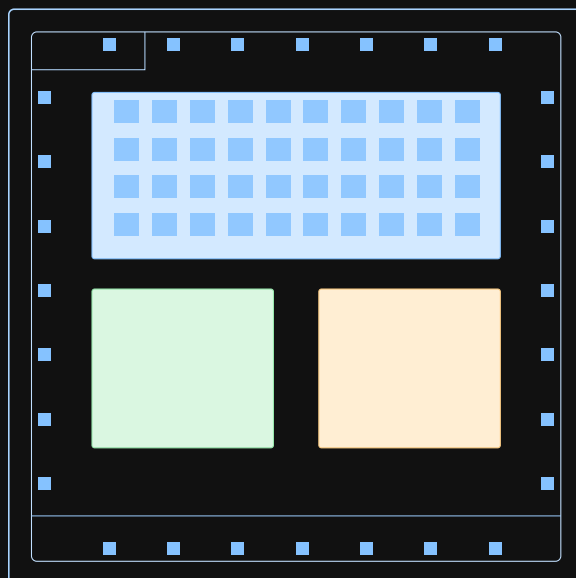


MNNW

MONTERREY NANOSCALE AND NUMERICAL WORKS

Penumbra

ILT on Silicon



TECHNICAL OVERVIEW · SEPTEMBER 2026

From	Roberto Treviño Cervantes, Founder
Subject	Hardware-Accelerated ILT — Capabilities Overview
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1. Context

The open problem is not whether ILT produces better masks — it is the cost of getting there. Full-chip ILT remains gated by the compute, time, and energy that rigorous mask optimization demands, which is what keeps aggressive ILT a premium step rather than a default one. **Penumbra** attacks that cost from the hardware side: a dedicated accelerator — today deployed on FPGAs — that produces curvilinear masks at a fraction of the energy and compute a CPU- or GPU-based flow would spend.

2. Penumbra ILT Core

71 %

LESS ENERGY

3.5×MORE EFFICIENT THAN A
GPU**15×**MORE EFFICIENT THAN A
CPU**8.68 ms**

PER TILE ON SILICON

The *Penumbra ILT Core* is a specialized processor that turns a target layout into a manufacturing-robust mask. The complete pipeline runs **self-contained on a single scalable core** — there is no off-chip traffic in the optimization loop.

Results on Real Hardware

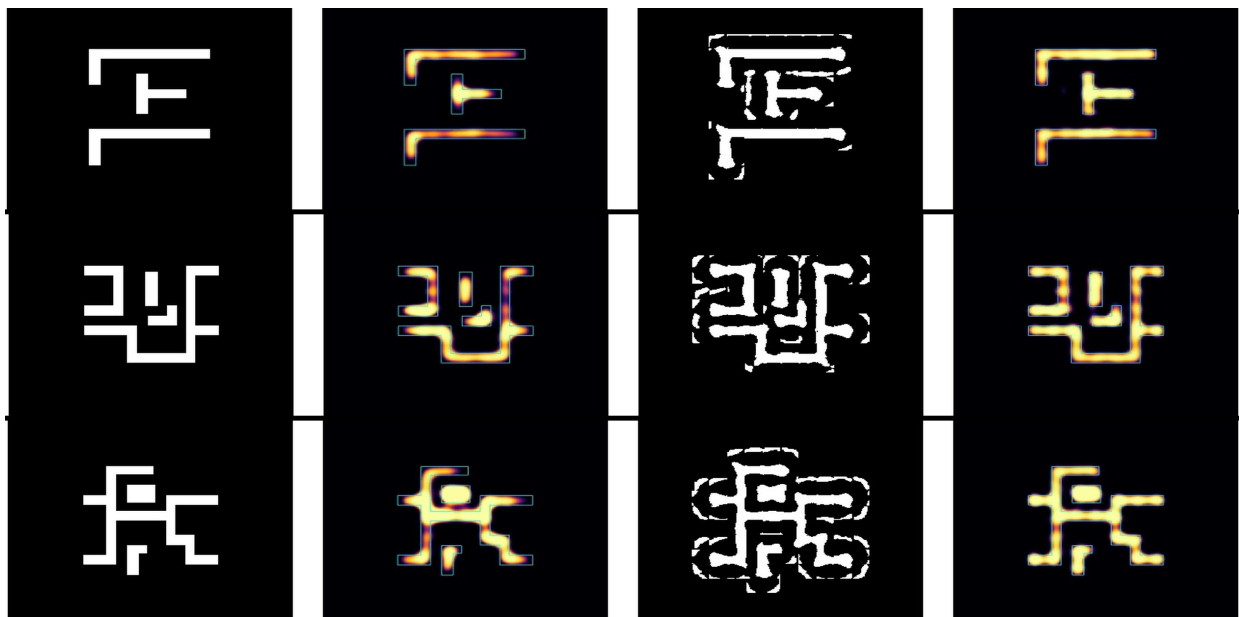


Figure 1. Curvilinear mask fragments generated on an FPGA by a single *Penumbra ILT Core* from LithoBench benchmark target patterns.

3. Verified on Silicon

Penumbra is currently distributed **on demand** through AWS F2 instances. It occupies only a modest fraction of a single FPGA, leaving comfortable headroom for **horizontal scaling** to full-chip throughput. We are currently working with **local academic institutions** to build a dedicated data-center around our coprocessor, supplying affordable **EDA compute** to firms looking to **nearshore** mask-optimization workloads — in step with the current US–Mexico semiconductor collaboration agenda.